

ASSESSMENT CO3 AT2

Fixed-Point vs Floating-Point Beamforming

Aim:

To simulate beamforming using fixed-point and floating-point arithmetic for an 8-element Uniform Linear Array (ULA) operating at 28 GHz and compare their performance in terms of beam pattern, quantization error, and computational requirements.

Objective

- To implement floating-point beamforming.
- To implement fixed-point beamforming with quantization.
- To compare the beam patterns of both implementations.
- To measure the quantization error between floating-point and fixed-point results.
- To analyze the computational requirements of both methods.
- To recommend the preferred implementation for practical applications.
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Procedure

1. Initialize the simulation parameters such as signal frequency (28 GHz), wavelength, element spacing, number of antenna elements, and scan angles.
2. Generate the steering vector for the desired beam direction using floating-point arithmetic.
3. Compute the beam pattern for the floating-point implementation.
4. Quantize the steering vector to obtain the fixed-point implementation.
5. Compute the beam pattern using the fixed-point steering vector.
6. Plot and compare the floating-point and fixed-point beam patterns.
7. Calculate the quantization error, including Mean Absolute Error (MAE), Maximum Error, and Root Mean Square Error (RMSE).
8. Compare the computational requirements of floating-point and fixed-point arithmetic.
9. Analyze the results and recommend the most suitable implementation based on accuracy and computational efficiency.

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%% Fixed-Point vs Floating-Point Beamforming

clc; clear; close all;

% Parameters

c=3e8; f=28e9; lambda=c/f; d=lambda/2;

N=8; theta0=30; theta=-90:0.2:90;

n=0:N-1;

%% Floating-Point Beamforming

w=exp(-1j*2*pi*d*n*sind(theta0)/lambda);

AFf=zeros(size(theta));

for k=1:length(theta)

    a=exp(-1j*2*pi*d*n*sind(theta(k))/lambda);

    AFf(k)=abs(w'*a);

end

AFf=AFf/max(AFf);

%% Fixed-Point Beamforming

scale=2^10;

wf=round(real(w)*scale)/scale + 1j*round(imag(w)*scale)/scale;

AFx=zeros(size(theta));

for k=1:length(theta)

    a=exp(-1j*2*pi*d*n*sind(theta(k))/lambda);

    af=round(real(a)*scale)/scale + 1j*round(imag(a)*scale)/scale;

    AFx(k)=abs(wf'*af);

end

AFx=AFx/max(AFx);

%% Beam Pattern

figure;

plot(theta,20*log10(AFf),'b','LineWidth',2); hold on;

plot(theta,20*log10(AFx),'r--','LineWidth',2);

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grid on;
xlabel('Angle (Degrees)');
ylabel('Normalized Gain (dB)');
title('Floating-Point vs Fixed-Point Beamforming');
legend('Floating Point','Fixed Point');
ylim([-50 5]);

%% Quantization Error
err=abs(AFf-AFx);

figure;
plot(theta,err,'k','LineWidth',2);

grid on;
xlabel('Angle (Degrees)');
ylabel('Magnitude Error');
title('Quantization Error');

%% Error Statistics
MAE=mean(err);
MAXE=max(err);
RMSE=sqrt(mean(err.^2));

fprintf('\n=== Quantization Error ===\n');
fprintf('Mean Absolute Error = %.8f\n',MAE);
fprintf('Maximum Error = %.8f\n',MAXE);
fprintf('RMSE = %.8f\n',RMSE);

fprintf('\n=== Computational Comparison ===\n');
fprintf('Floating Point: High precision, higher memory, higher computational cost.\n');
fprintf('Fixed Point: Lower precision, lower memory, faster hardware implementation.\n');
fprintf('\n=== Recommendation ===\n');
if RMSE<0.01
    fprintf('Fixed-point implementation provides nearly identical performance.\nPreferred for FPGA/ASIC and real-time communication systems.\n');
else
    fprintf('Floating-point implementation is preferred due to lower quantization error.\n');
end

```

Command Window Output

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Command Window

=== Quantization Error ===
Mean Absolute Error = 0.00007631
Maximum Error = 0.00032438
RMSE = 0.00009537

=== Computational Comparison ===
Floating Point: High precision, higher memory, higher computational cost.
Fixed Point: Lower precision, lower memory, faster hardware implementation.

=== Recommendation ===
Fixed-point implementation provides nearly identical performance.
Preferred for FPGA/ASIC and real-time communication systems.
>>
```

Figure 1: Beam Pattern

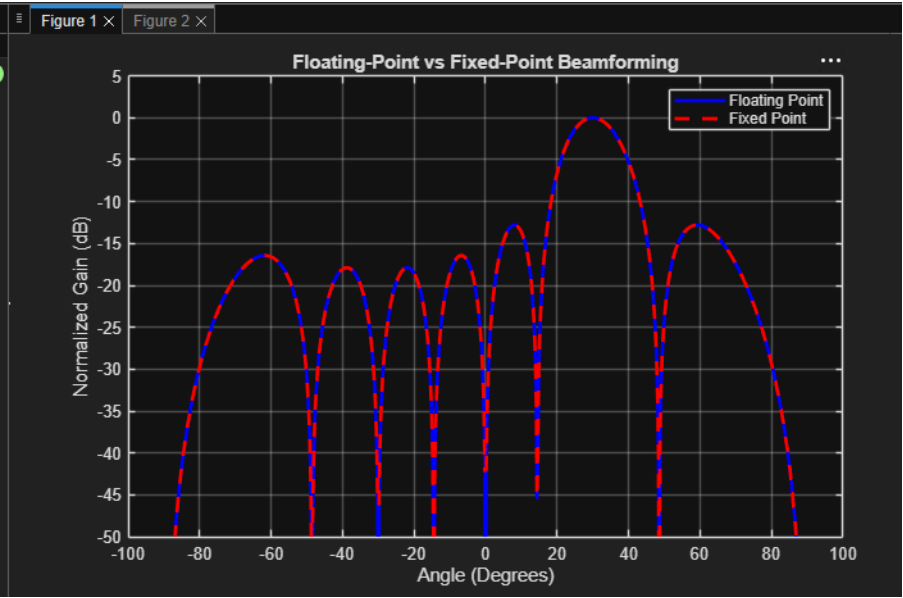
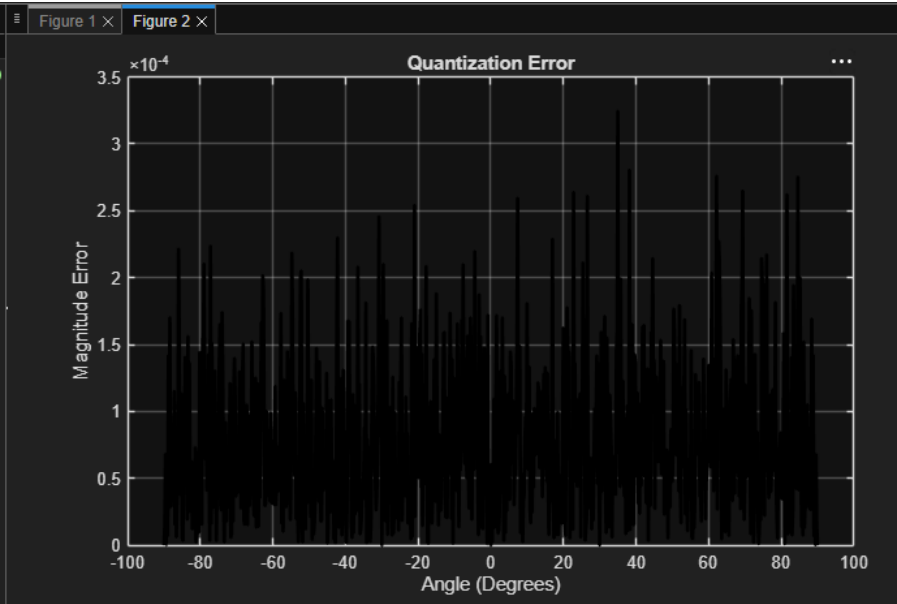


Figure 2: Quantization Error



Result

The floating-point and fixed-point beam patterns are nearly identical.

Quantization errors are very small:

- MAE = 0.00007631
- Maximum Error = 0.00032438
- RMSE = 0.00009537

Hence, the fixed-point implementation is recommended for FPGA/ASIC and real-time communication systems due to its lower computational complexity and memory usage while maintaining comparable beamforming performance.